

Flynn, Nikolakoudis & Sastry (2026): Pricing and Production Without The Invisible Hand

New & Old Ideas in Macroeconomics

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Overview

Motivation:

- ▶ Perfect competition: only quantities are determined by agents
 - Arrow (1959): no rational decision on prices
- ▶ Imperfect competition: only prices are determined by agents
 - Firms need to produce , even if they would rather not do so

⇒ Delegation of pricing or production to **invisible hand** undesirable

This paper:

- ▶ Equilibrium determination of pricing and production BOTH made by agents

Overview

Implications:

- ▶ Fully characterized GE outcomes
- ▶ Markups given by inverse probability of rationated demand (vs Lerner, 1934)
- ▶ Increases in volatility σ_U have **first-order** effects
- ▶ Aggregate demand externalities can turn negative
 - $\uparrow$ production in rationed markets $\implies \downarrow$ demand in non-rationed markets
- ▶ Uncertainty shocks $\implies \uparrow$ prices $\implies \downarrow$ consumption + $\leftrightarrow$ employment
- ▶ Monetary non-neutrality: $\downarrow$ with σ_U , discontinuous for $\sigma_U \rightarrow 0$, $\uparrow\downarrow$ mkt power
- ▶ Unanticipate monetary shocks raise labor productivity

Overview

Where It Sits In the Literature

Barro–Grossman (1971)

- ▶ Fixprice disequilibrium in goods and labor markets.
- ▶ Effective demands depend on rationing in other markets.
- ▶ Regimes: Keynesian unemployment, classical unemployment, repressed inflation.

Bénassy (1993)

- ▶ General theory of rationing schemes and quantity signals.
- ▶ Voluntary exchange: no agent trades more than desired.
- ▶ Efficient exchange: no mutually desired trades are left unrealized.

Michaillat–Saez (2015)

- ▶ Keep Barro–Grossman architecture.
- ▶ Replace binary rationing regimes with continuous market tightness.
- ▶ Aggregate demand affects unemployment through product-market idle time.

Common object: markets need not clear, and quantity constraints generate cross-market spillovers.

Common difficulty: with several markets, equilibrium is regime-by-regime: which side is rationed in which market?

Overview

Why Uncertainty?

- **Classical disequilibrium** keeps the short-side rule:

$$q_i = \min\{q_i^s, q_i^d\}.$$

But the economy must be solved by cases: slack/rationed goods markets, slack/rationed labor markets, and spillovers across them.

- **Matching functions** make the problem smooth:

$$q_i = m(q_i^s, q_i^d) \quad \text{with} \quad m(q_i^s, q_i^d) \leq \min\{q_i^s, q_i^d\}.$$

This delivers a continuous tightness state, but generically allows $m(q_i^s, q_i^d) < \min\{q_i^s, q_i^d\}$: mutually beneficial trades can fail.

- **This paper** keeps voluntary and efficient exchange, but firms choose p_i, L_i before the imbalance $U_i = A_i/z_i$ is realized:

$$q_i = \min\{A_i L_i, z_i p_i^{-\eta}\}, \quad t_i = \frac{p_i^{-\eta}}{L_i}, \quad r = \mathbb{P}(t_i \geq U_i).$$

Uncertainty turns regimes into **probabilities/shares of markets**: rationed if $t_i \geq U_i$, slack if $t_i < U_i$, Walrasian only on a probability-zero threshold. The kink from efficient exchange is exactly why σ_U has first-order effects.

Overview

Plan of the presentation:

1. Pricing and production in Partial Equilibrium
2. A static General Equilibrium model
3. Pricing and production in General Equilibrium
4. Dynamics: A New Old Keynesian model

Pricing and Production in PE

Model: The Firm's Problem

$$\begin{aligned} & \sup_{p,L} \int_{\mathbb{R}_4^{++}} \Lambda(pq - wL) dG(\Lambda, A, z, w) \\ \text{s.t.} \quad & q = \min\{q^s, q^d\}, \quad q^s = AL, \quad q^d = zp^{-\eta} \end{aligned}$$

where (Λ, A, z, w) are jointly normally distributed with μ, Σ and G .

► Two principles (Bénassy, 1993):

- **Voluntary exchange**: no trade can be forced: $q \leq q^s$ and $q \leq q^d$
- **Efficient exchange**: no trade will fail if desired: never that both $q < q^s$ and $q < q^d$

Consequences:

$$q = \min\{q^s, q^d\}$$

Pricing and Production in PE

The Prevalence of Walrasian Disequilibrium

$$\sup_{p,L} \int_{\mathbb{R}_4^{++}} \Lambda(\min\{zp^{1-\eta}, pAL\} - wL) dG(\Lambda, A, z, w)$$

where (Λ, A, z, w) are jointly normally distributed with μ, Σ and G .

$$q = \min\{q^s, q^d\}$$

► Three possible outcomes:

- $q^s > q^d$: **Slack** market
- $q^s < q^d$: **Rationed** market
- $q^s = q^d$: **Walrasian equilibrium** (kink in the revenue function)

Pricing and Production in PE

Optimal Firm Behaviour

Theorem 1

There exists a unique price p^ and labor input L^* . Moreover, optimal tightness $t^* = \frac{p^{*-\eta}}{L^*}$ is the unique solution to:*

$$t = \frac{1}{\eta - 1} \frac{\mathbb{E}[\Lambda A \mathbb{I}[t \geq U]]}{\mathbb{E}[\Lambda z \mathbb{I}[t < U]]}$$

where $U = A/z$. Moreover, the optimal price is given by:

$$p^* = \frac{\mathbb{E}[\Lambda w]}{\mathbb{E}[\Lambda A \mathbb{I}[t^* \geq U]]}$$

and the optimal labor input is then given by $L^ = \frac{p^{*-\eta}}{t^*}$*

Pricing and Production in PE

Optimal Firm Behaviour

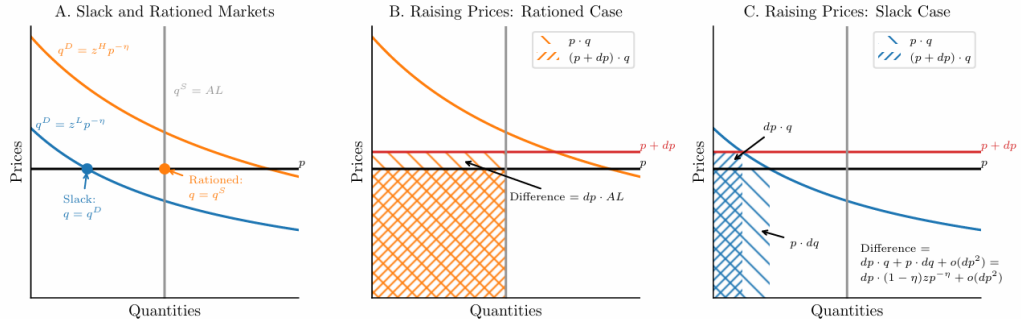


Figure: The Incentives to Raise Prices

Pricing and Production in PE

Optimal Tightness t^*

$$\text{Tightness : } t := \frac{p^{-\eta}}{L}$$

► Part of demand/supply the firm can choose:

- $\downarrow p \implies \uparrow p^{-\eta} \implies \uparrow t$ vs $\downarrow L \implies \uparrow t$

$$\text{Shock to supply/demand : } U := \frac{A}{z}$$

► Part of demand/supply the firm cannot choose:

- Ex post: $\uparrow z \implies \downarrow U \implies \uparrow \mathbb{P}(t \geq U)$
- Ex ante: if $\mathbb{E}[\uparrow z] \implies \downarrow t^*$
- Firm wants to **balance** the two effects

Pricing and Production in PE

Optimal Tightness t^*

Effect of an increase in prices p :

- ▶ If market is rationed:

$$\mathbb{E}[\Lambda A | t \geq U] dp \quad (\text{gains})$$

- ▶ If market is slack:

$$(\eta - 1) \mathbb{E}[\Lambda z p^{-\eta} | t < U] dp \quad (\text{losses})$$

- ▶ Walrasian equilibrium has 0 probability.

Optimal tightness t^* balances:

$$(\eta - 1) \mathbb{E}[\Lambda z p^{-\eta} | t < U] \times \mathbb{P}(t < U) = \mathbb{E}[\Lambda A | t \geq U] \times \mathbb{P}(t \geq U)$$

Pricing and Production in PE

Prices and Markups without the Invisible Hand

- ▶ From the first order condition with respect to *inputs*
- ▶ Increasing inputs:

$$\underbrace{\mathbb{E}[\Lambda A p | t \geq U]}_{\text{market is rationed}} \mathbb{P}(t \geq U) + \underbrace{0}_{\text{market is slack}} \mathbb{P}(t < U) \quad \text{vs} \quad \underbrace{\mathbb{E}[\Lambda w]}_{\text{cost of inputs}}$$

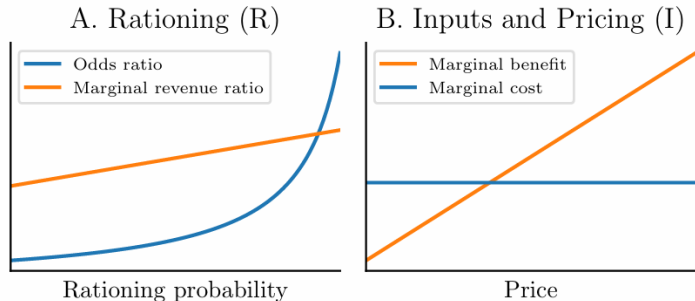
At the optimal price p^* , the firm is indifferent between increasing and decreasing inputs:

$$p^* = \frac{1}{r} \frac{\mathbb{E}[\Lambda w]}{\mathbb{E}[\Lambda A | t^* \geq U]} \quad \text{where } r = \mathbb{P}(t \geq U)$$

- ▶ Markup is given by the inverse of the probability of being rationed
 - Where is Lerner (1934) formula?

Pricing and Production in PE

Block Recursivity



$$\frac{r}{1-r} = \frac{\mathbb{E}[\Lambda z | T(r) \leq U]}{\mathbb{E}[\Lambda A | T(r) \geq U]} \quad (R)$$

$$\mathbb{E}[\Lambda w] = r \mathbb{E}[\Lambda A | T(r) \geq U] p \quad (I)$$

with $T(r)$ a function of the probability of rationing r .

The Effect of Uncertainty

The Complete Information Limit

Proposition 1

In the complete information limit:

1. Prices are a constant taste-based markup over nominal marginal costs:

$$p^{FI} \equiv \lim_{\delta \rightarrow 0} p^* = \frac{\eta}{\eta - 1} \exp\{\mu_w - \mu_A\}.$$

2. Labor converges to its full information value:

$$L^{FI} \equiv \lim_{\delta \rightarrow 0} L^* = \left(\frac{\eta}{\eta - 1} \right)^{-\eta} \exp\{\mu_z + (\eta - 1)\mu_A - \eta\mu_w\}.$$

3. The supply-demand imbalance in choices is the fundamental supply-demand imbalance:

$$t^{FI} \equiv \lim_{\delta \rightarrow 0} t^* = \exp\{\mu_A - \mu_z\}.$$

4. The rationing probability $r = \mathbb{P}[q^d \geq q^s]$ is the inverse taste-based markup:

$$\lim_{\delta \rightarrow 0} r^* = \frac{\eta - 1}{\eta}.$$

The Effect of Uncertainty

The Complete Information Limit

- ▶ Even with infinitesimal uncertainty, rationing probability is non-zero
- ▶ Why $r^* = \frac{\eta-1}{\eta}$? $(\eta - 1) \times (1 - \mathbb{P}[t \geq U]) = 1 \times \mathbb{P}[t \geq U]$
 - Lerner (1934) formula is back.
 - Perfect competition ($\eta \rightarrow \infty$) $\implies r \rightarrow 1$
 - Monopoly ($\eta \rightarrow 1$) $\implies r \rightarrow 0$
- ▶ Uncertainty is the key element!

The Effect of Uncertainty

The First-Order Effect of Volatility

Proposition 2

Up to a first-order approximation in volatility, the firm's optimal price and labor inputs are:

$$\begin{aligned}\ln p^* &= \ln p^{FI} + \left(\phi(\kappa) - \frac{\kappa}{\eta} \right) \sigma_U, \\ \ln L^* &= \ln L^{FI} - \eta \phi(\kappa) \sigma_U,\end{aligned}$$

where $\kappa = \Phi^{-1} \left(\frac{\eta-1}{\eta} \right)$.

Moreover, the rationing probability is:

$$r = \frac{\eta-1}{\eta} \left[1 + \left(\frac{\kappa}{\eta} + \left(\frac{\beta^S}{\eta-1} + \beta^D \right) \phi(\kappa) \right) \sigma_U \right].$$

Why first order effects? Kink in the payoff function

The Effect of Uncertainty

How Uncertainty Affects Rationing, Price and Labor

Rationing:

- ▶ κ/η term: mechanical effect of volatility on marginal revenue ratio;
 - if $\kappa < 0$: $\uparrow \sigma_U \Rightarrow \downarrow$ marginal revenue ratio (slack market less likely).
- ▶ $\left(\frac{\beta^S}{\eta-1} + \beta^D\right) \phi(\kappa)$ term: inferences about z (slack mkt) and A (rationed mkt);
 - low z in slack states $\Rightarrow$ firm $\downarrow$ tightness ($\uparrow p$) $\Rightarrow \downarrow$ probability of rationing.

Price and labor:

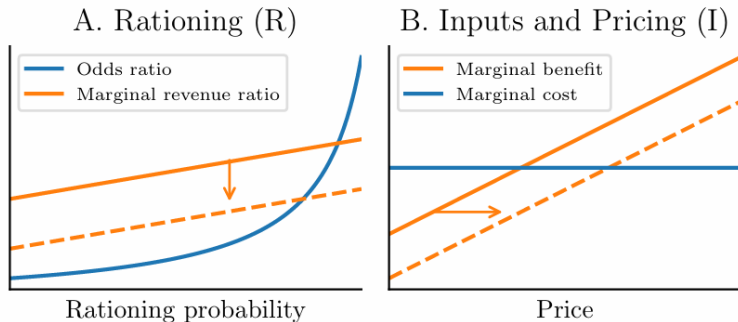
- ▶ Benefit of hiring only in rationed states.
- ▶ Price charged moves in opposite direction of rationing probability.

The Effect of Uncertainty

Firms Shrink When They Are Uncertain

Corollary 2

The presence of imperfect information reduces hiring, $\ln L < \ln L^{FI}$, and increase prices $\ln p > \ln p^{FI}$.



The Effect of Uncertainty

Uncertainty Based Markups

$$\ln \mathcal{M} = \underbrace{\ln \left(\frac{\eta}{\eta - 1} \right)}_{\text{Taste-Based Markup}} + \underbrace{\left(\phi(\kappa) - \frac{\kappa}{\eta} \right)}_{\text{Uncertainty-Based Markup}} \sigma_U$$

Rationalize:

- ▶ Demand side approach of estimating markups - elasticity of demand
- ▶ Supply side approach of estimating markups - marginal costs
- ▶ Markup shocks in business cycle analysis

A Static General Equilibrium Model

The Household's Problem

Goal: how uncertainty shocks and monetary shocks propagate.

Setup:

$$C = \left(\int \theta_i^{\frac{1}{\eta}} c_i^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}} \quad \text{CES aggregator, } \eta > 1$$

$$U(C, M/P, N) = \ln C + \ln \left(\frac{M}{P} \right) - \alpha N \quad \text{Utility function}$$

$$PC + M = wN + \Pi + T \quad \text{Budget constraint}$$

where θ_i variety-specific taste shock (IID log-normal $\mu_\theta, \sigma_\theta^2$)

Timing:

- ▶ Households chooses labor supply N when firms choose labor demand L .
- ▶ Households choose consumption C after firms has produced.

A Static General Equilibrium Model

Optimal Household Behavior

Given expenditure E , households choose varieties subject to firms' produced quantities $\bar{c}_i$:

$$\max_{\{c_i\}_{i \in [0,1]}} \left(\int_0^1 \theta_i^{\frac{1}{\eta}} c_i^{\frac{\eta-1}{\eta}} di \right)^{\frac{\eta}{\eta-1}} \quad \text{s.t.} \quad \int_0^1 p_i c_i di = E, \quad c_i \leq \bar{c}_i \quad \forall i \in [0, 1]$$

Rank varieties by their supply-demand imbalance:

$$\iota_i = \frac{p_i^{-\eta} / L_i}{A_i / \theta_i} = \frac{t_i}{U_i}$$

- ▶ Low ι_i : relatively scarce goods, likely to be rationed.
- ▶ High ι_i : relatively abundant goods, likely to be consumed at capacity.

A Static General Equilibrium Model

Optimal Household Behavior

Proposition 6

There exists a unique threshold $k \in [0, 1]$ such that

$$\begin{aligned} c_j &= \frac{E - \int_k^1 p_i \bar{c}_i di}{\int_0^k \theta_i p_i^{1-\eta} di} \theta_j p_j^{-\eta} && \text{for all } j \in [0, k), \\ c_j &= \bar{c}_j && \text{for all } j \in (k, 1]. \end{aligned}$$

Moreover, k is the unique solution to the first condition evaluated at $j = k$, and

$$C = \left[\left(E - \int_k^1 p_i \bar{c}_i di \right)^{\frac{\eta-1}{\eta}} \left(\int_0^k \theta_i p_i^{1-\eta} di \right)^{\frac{1}{\eta}} + \int_k^1 \theta_i^{\frac{1}{\eta}} \bar{c}_i^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}}.$$

A Static General Equilibrium Model

Cross-Substitution Without the Invisible Hand

- ▶ If all the markets are slack ($k = 1$) (Dixit and Stiglitz, 1977):

$$c_j = \theta_j \frac{E}{P} \left(\frac{p_j}{P} \right)^{-\eta}, \quad P = \left(\int_0^1 \theta_i p_i^{1-\eta} di \right)^{\frac{1}{1-\eta}}$$

- ▶ With rationing ($k < 1$):

$$c_j = \frac{E - \int_k^1 p_i \bar{c}_i di}{P_k} \left(\frac{p_j}{P_k} \right)^{-\eta}, \quad P_k = \left(\int_0^k \theta_i p_i^{1-\eta} di \right)^{\frac{1}{1-\eta}}$$

Cross-substitution: $\uparrow t_i$ by $\downarrow \bar{c}_i \implies$ positive demand shifter for slack markets.

- ▶ **Characterization of spillover effects**

A Static General Equilibrium Model

Labor Supply Decisions

- ▶ FOCs:

$$w \frac{1}{PC} \quad \text{and} \quad \frac{1}{PC} = \frac{1}{M}$$

- ▶ Money holdings

$$w = \alpha M$$

- ▶ Nominal consumption equal to money holdings:

$$E = PC = M$$

A Static General Equilibrium Model

Equilibrium

Timing:

1. Morning:
 - Government set money supply $\bar{M}$
 - Firms set prices p_i and hire labor L_i .
 - Household chooses labor supply N .
2. Afternoon:
 - Production takes place, $\bar{c}_i = A_i L_i$.
3. Evening:
 - Households choose money holding M and consume $c_i \forall i \in [0, 1]$.

Simplification:

- ▶ labor market clears in the morning, so $N = \int_0^1 L_i di$.
- ▶ money market clears in the evening, so $M = T = \bar{M}$.

Note: Only failure of the market clearing in the goods market

A Static General Equilibrium Model

Equilibrium

Variables: $\{\{t_i, p_i, L_i, c_i, A_i, \theta_i, z_i\}_{i \in [0,1]}, C, M, P, E, \Pi, T, N, k, w, \bar{M}, \Lambda\}$.

1. Firm optimality

Each firm chooses price, labor, and tightness, taking z_i as given.

$$\begin{aligned}t_i &= \frac{1}{\eta - 1} \frac{\mathbb{E}[\Lambda A_i \mathbb{I}[t_i \geq U_i]]}{\mathbb{E}[\Lambda z_i \mathbb{I}[t_i \leq U_i]]}, \\U_i &= \frac{A_i}{z_i}, \\p_i &= \frac{\mathbb{E}[\Lambda w]}{\mathbb{E}[\Lambda A_i \mathbb{I}[t_i \geq U_i]]}, \\L_i &= \frac{p_i^{-\eta}}{t_i}.\end{aligned}$$

2. Household demand

Consumption respects rationing; k separates rationed from slack markets.

$$\begin{aligned}c_i &= \min\{z_i p_i^{-\eta}, \bar{c}_i\}, \\\bar{c}_i &= A_i L_i, \\z_i &= D \theta_i, \\D &= \frac{E - \int_k^1 p_i \bar{c}_i di}{\int_0^k \theta_i p_i^{1-\eta} di}, \\\bar{c}_k &= \frac{E - \int_k^1 p_i \bar{c}_i di}{\int_0^k \theta_i p_i^{1-\eta} di} \theta_k p_k^{-\eta}.\end{aligned}$$

A Static General Equilibrium Model

Equilibrium

3. Aggregate household choices: consumption combines slack-market demand and rationed-market capacity.

$$C = \left[\left(E - \int_k^1 p_i \bar{c}_i di \right)^{\frac{\eta-1}{\eta}} \left(\int_0^k \theta_i p_i^{1-\eta} di \right)^{\frac{1}{\eta}} + \int_k^1 \theta_i^{\frac{1}{\eta}} \bar{c}_i^{\frac{\eta-1}{\eta}} di \right]^{\frac{\eta}{\eta-1}}. \quad w = \alpha M, \quad E = PC = M.$$

4. Clearing and accounting: money and labor clear, profits are rebated, and prices/SDF are consistent with consumption.

$$M = T = \bar{M}, \quad N = \int_0^1 L_i di, \quad \Pi = \int_0^1 (p_i c_i - w L_i) di, \quad P = \frac{\int_0^1 p_i c_i di}{C}, \quad \Lambda = \frac{1}{PC}.$$

Pricing and Production in GE

Characterizing Equilibrium

Aim: Study propagation of monetary shocks ($\bar{M}\Delta$) and uncertainty shocks (σ_U).

Theorem 2

Up to a first-order approximation in volatility, the unique equilibrium level of mean demand in non-rationed markets is

$$\mu_z = \ln \left[\left(\frac{\eta\alpha}{(\eta-1)e^{\mu_A}} \right)^{\eta-1} \bar{M}^\eta \right] + \eta\phi(\kappa)\sigma_U.$$

Thus, equilibrium prices, labor input, and aggregate consumption are

$$\ln p = \ln \left(\frac{\eta\alpha\bar{M}}{(\eta-1)e^{\mu_A}} \right) + \left(\phi(\kappa) - \frac{\kappa}{\eta} \right) \sigma_U,$$

$$\ln L = \ln \left(\frac{\eta-1}{\eta\alpha} \right),$$

$$\ln C = \ln \left(\frac{(\eta-1)e^{\mu_A + \frac{\mu_\theta}{\eta-1}}}{\eta\alpha} \right) - \left(\phi(\kappa) + \frac{\kappa}{\eta} \right) \sigma_U.$$

Pricing and Production in GE

Characterizing Equilibrium

- ▶ $\sigma_U \rightarrow 0$: standard monopolistic benchmark with market clearing.
- ▶ Prices in GE and PE coincides (depends only on rationing probability).
- ▶ Labor in GE depends on spillover:
 - PE effect (insurance) exactly offset GE effect (spillovers)
- ▶ Aggregate consumption is decreasing in uncertainty
 - $\uparrow \sigma_U \Rightarrow \uparrow P; M = PC \Rightarrow \downarrow C$.

Pricing and Production in GE

The Effects of Uncertainty Shocks

- Propagation as markup shocks to price and consumption.

$$\ln p = \underbrace{\ln \left(\frac{\alpha M}{e^{\mu_A}} \right)}_{\text{efficient prices}} + \underbrace{\ln \mathcal{M}}_{\text{markup}}, \quad \ln C = \underbrace{\ln \left(\frac{e^{\mu_A + \frac{\mu_\theta}{\eta - 1}}}{\alpha} \right)}_{\text{efficient consumption}} - \underbrace{\ln \mathcal{M}}_{\text{markup}}$$

where:

$$\ln \mathcal{M} = \underbrace{\ln \left(\frac{\eta}{\eta - 1} \right)}_{\text{taste-based markup}} + \underbrace{\left(\phi(\kappa) - \frac{\kappa}{\eta} \right) \sigma_U}_{\text{uncertainty-based markup}}$$

- Depresses aggregate labor productivity:

$$\ln \frac{C}{L} = \underbrace{\mu_A + \frac{\mu_\theta}{\eta - 1}}_{\text{steady-state level}} - \underbrace{\left(\phi(\kappa) + \frac{\kappa}{\eta} \right) \sigma_U}_{\text{effect of uncertainty}}$$

Pricing and Production in GE

The Effects of (Unanticipated) Monetary Shocks

- ▶ Unanticipated monetary shocks have real effects
 - Goods market doesn't clear *ex-post*
- ▶ Monetary non-neutrality:

$$\mathcal{M} = \left. \frac{\partial \ln C}{\partial \ln \Delta} \right|_{\ln \Delta = 0}$$

Theorem 3

Up to first-order approximation in volatility, the extent of monetary non-neutrality is given by:

$$\mathcal{M} = 1 - \frac{1}{\eta} \left(\phi(k) + \frac{\eta - 1}{\eta} \kappa \right) \sigma_U$$

Pricing and Production in GE

The Effects of (Unanticipated) Monetary Shocks

$$\frac{\partial \ln C}{\partial \ln \Delta} = \underbrace{\left(\frac{\int_0^1 c_i di}{C} \right)^{\frac{\eta-1}{\eta}}}_{\text{consumption misallocation}} \times \underbrace{\eta \int_0^k \left(\frac{\int_0^1 c_i di}{c_i} \right)^{\frac{1}{\eta}} di}_{\text{average excess consumption in slack markets}}$$

- Passthrough depends on:
 - consumption misallocation (matters only to second order)
 - how much more consumption is in slack markets relative to average market

Pricing and Production in GE

The Role of Uncertainty

- ▶ Uncertainty and slack are the key element for monetary non-neutrality
- ▶ It boils down to the kink in the payoff function
- ▶ Degree of monetary non-neutrality is determined by market power:
 - Non-monothonic relation driven by:
 1. low market power $\Rightarrow$ $\uparrow$ cost of overproduction $\Rightarrow$ $\uparrow$ probability of being rationed
 2. low market power $\Rightarrow$ $\downarrow$ welfare loss from overproduced varieties

Dynamics: A New Old Keynesian Model

The Setup

- ▶ Dynamic setting
- ▶ Households
- ▶ Calvo (1983) price setting
- ▶ Firms' prices and quantities are both sticky
⇒ Goods markets do not necessarily clear in each period

Key feature: Walrasian disequilibrium is not assumed!

Dynamics: A New Old Keynesian Model

Linearized NOK Model

Proposition 8

When log-linearized around the steady state, and given processes for monetary shocks and volatility shocks $\{\hat{M}_\tau, \hat{\sigma}_{U,\tau}\}_{\tau \in \mathbb{N}}$, dynamic equilibria solve:

1. **New Keynesian Phillips Curve**

$$\pi_\tau = \Gamma(\hat{M}_\tau - \bar{p}_\tau) + \beta \mathbb{E}_\tau[\pi_{\tau+1}] + \zeta \hat{\sigma}_{U,\tau}.$$

2. **New Old Keynesian Tightness Curve**

$$v_\tau = -\Gamma(\hat{D}_\tau + \bar{t}_\tau) + \beta \mathbb{E}_\tau[v_{\tau+1}] + \psi \hat{\sigma}_{U,\tau}.$$

3. **Demand Spillovers Curve**

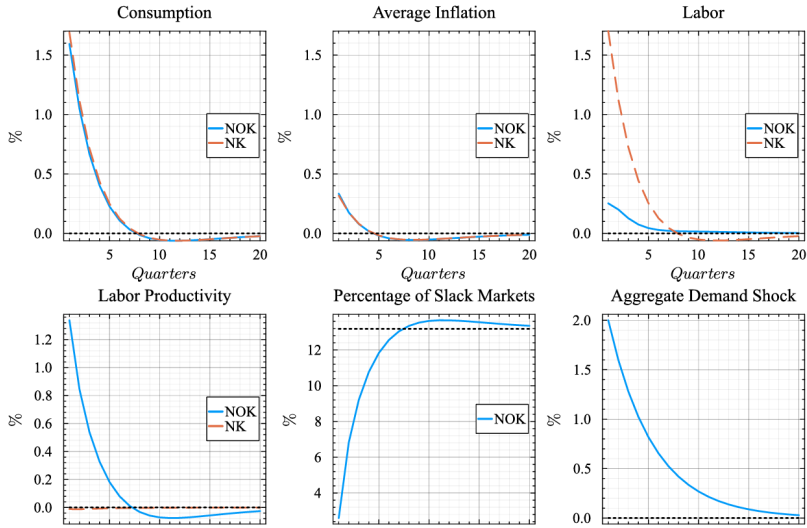
$$\hat{D}_\tau = \gamma_M \hat{M}_\tau + \gamma_p \bar{p}_\tau + \gamma_t \bar{t}_\tau + \gamma_U \hat{\sigma}_{U,\tau}.$$

where $\Gamma = \frac{(1-\omega)(1-\beta\omega)}{\omega}$ and $(\zeta, \psi, \gamma_M, \gamma_p, \gamma_t, \gamma_U)$ are constants.

- ▶ Propagation of shocks: as in standard NK model
- ▶ Mechanics: depends on coevolution of mkt tightness and spillovers

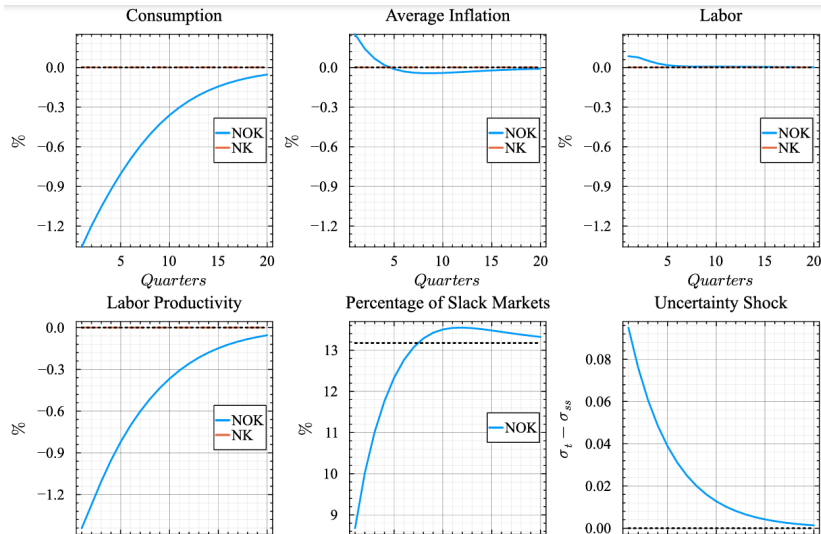
Dynamics: A New Old Keynesian Model

The Propagation of Monetary Shocks



Dynamics: A New Old Keynesian Model

The Propagation of Uncertainty Shocks



Conclusions

- ▶ GE model with prices and quantities determined by rational agents
- ▶ Micro level: Lerner condition is just a special case with no uncertainty
 - Uncertainty has first-order effects, increase prices and depress labor demand
- ▶ Macro level: well defined distribution of rationed and slack markets
- ▶ Characterization of spillover effects in CES demand case
- ▶ New Old Keynesian model:
 - Monetary shocks propagate as in standard NK model, different channel
 - Uncertainty shocks propagate as markup shocks